LLM WatchTower: Safeguarding Chat Interfaces from Malicious Intent

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Abstract

The proliferation of Large Language Models (LLMs) has transformed chat interfaces, facilitating human-bot interactions with remarkable problem-solving skills. This advancement also brings with it an increased vulnerability to malicious actors. The crux of the concern lies in the models' inclination to be helpful, without having a deep understanding of the intentions behind the requests or the potential implications of the information they provide. This lack of discernment can lead to scenarios where LLMs inadvertently produce harmful, biased, or sensitive content, or are manipulated through carefully crafted inputs to bypass safeguards—a practice known as "prompt injection" or "jailbreaking." This paper explores the enhancement of moderation capabilities in smaller LLMs, specifically through the incorporation of prompt tuning with causal reasoning. Utilizing models like Mistral-7B, this study innovates by not only classifying texts as malicious or benign but also by elaborating on the reasoning behind these classifications, providing a dual-layer safeguarding mechanism. Our findings reveal that our fine-tuned model, WatchTower, outperforms existing solutions in terms of Recall and F1 score on the balanced ToxicChat dataset. While the inclusion of causal reasoning did not significantly enhance the already saturated model performance, it introduced an element of model interpretability with no performance drawbacks. This enhancement can be valuable in real-world applications, as it bolsters trust and transparency in AI decision-making, potentially helping to ensure compliance with regulations.

Introduction

AI-driven systems are at the forefront of delivering responsive, intuitive, and seemingly intelligent dialogue capabilities that greatly enhance user experience. However, the sophistication of these models also introduces new vulnerabilities, notably through the risks associated with prompt injections and jailbreak attempts, where malicious inputs aim to subvert the AI's operational guidelines. A notable instance that highlights the urgency for improved safeguards occurred recently when an AI-powered chatbot by General Motors inadvertently agreed to sell a Chevy Tahoe truck for merely $1 due to a crafted prompt that exploited underlying vulnerabilities in the language model's processing logic. Such incidents not only present substantial financial risks but also threaten the integrity and reliability of using LLMs in business operations, prompting an immediate need for more robust defense mechanisms.

The central hypothesis of this paper is that incorporating prompt tuning with causal reasoning into smaller LLMs can enhance the accuracy of content moderation and malicious prompt detection. Unlike conventional models that consider these tasks as a multilabel classification problem, integrating reasoning capabilities allows the model to not only identify harmful content but also provide a justifying rationale behind its classifications. This dual-layer approach aims to leverage LLMs' inherent ability to process logic and reasoning, thereby improving the accuracy of content moderation and offering insights into the decision-making processes of AI systems, which could further be leveraged to fine-tune the models.

To effectively address this hypothesis, it is crucial to establish a clear understanding of the underlying concepts:

1. **Prompt Injection and Jailbreak Attacks**: These are techniques used by malicious actors to manipulate AI models into violating their operational guidelines or producing undesirable outcomes. They exploit the model's tendency to generate helpful and coherent responses without a deep understanding of the implications or the intentions behind the input prompts.
2. **Causal Reasoning**: This involves the capability of an AI to comprehend and articulate the causality between events in a given text. By integrating causal reasoning, the model is expected to not just react to the prompts but to understand the 'why' behind them, which is crucial for distinguishing between benign and malicious intents.
3. **Prompt Tuning**: Refers to the fine-tuning process where LLMs are specifically adjusted to improve their response accuracy to certain types of queries. This study utilizes prompt tuning to teach the LLM

to recognize and reason about the causal dynamics in conversations, enhancing its preventive capabilities against sophisticated attacks.

The broader implications of this research extend beyond the immediate scope of content moderation. By demonstrating the potential for smaller LLMs to effectively detect and provide reasoning about malicious prompts, we open up new avenues for deploying these models in resource-constrained environments. Furthermore, the incorporation of causal reasoning aligns with the growing demand for explainable AI, contributing to the development of more transparent and accountable AI systems.

To test this hypothesis, we employ a multi-stage methodology. First, we curate a diverse dataset from multiple sources, balancing malicious and benign examples. Second, we augment this dataset with causal reasoning generated by GPT-3.5, explaining the rationale behind the malicious classification. Third, we conduct a series of training runs on smaller LLMs like Mistral-7B, comparing the performance of models trained solely on binary classification against those trained to predict causal reasoning alongside the malicious flag. Finally, we evaluate the best-performing models against popular community solutions like LlamaGuard and LLMGuard on the ToxicChat dataset, assessing their effectiveness in detecting various types of prompt injection attacks.

The findings confirm that while the inclusion of causal reasoning did not drastically change the quantitative outcomes in terms of detection accuracy (as both approaches yielded similar F1 scores), it significantly enhanced the model's interpretability and trustworthiness. This qualitative improvement is pivotal for real-world applications where justifications of AI decisions are crucial for user trust and regulatory compliance. Our enhanced model, WatchTower, demonstrated superior Recall (95.6%) and F_1 (88.6%) on the balanced ToxicChat compared to LlamaGuard-7B (Recall=61.5% & F1=71.3%) and Deberta based LLM-Guard (Recall=53.8% & F1=68.1%). We also validated that this performance gain is not attributed to any kind of data leakage by accessing the lexical similarity between our training data and ToxicChat data.

Related Work

The rapid advancements in Large Language Models (LLMs) have led to a surge in research exploring ways to safeguard these models against malicious attacks, particularly in the context of human-AI conversations. This section will discuss the most relevant work in this domain, focusing on their methodologies, innovations, and how they relate to our proposed approach.

Content Moderation and Risk Classification: Llama Guard [1] leverages an LLM-based model to classify potential safety risks using a comprehensive safety risk taxonomy. It employs instruction tuning on a carefully curated dataset reflecting real-world conversation dynamics, allowing it to outperform existing content moderation tools. This targeted approach aligns closely with our goal of understanding and mitigating risks in real-world scenarios. However, while Llama Guard focuses on multi-class classification and binary decision scores, our approach incorporates causal reasoning to provide justifications for the model's decisions, adding a layer of interpretability.

Intrinsic Defense Mechanisms: Defending ChatGPT against Jailbreak Attack via Self-Reminders [2] proposes embedding ethical guidelines directly into ChatGPT's response mechanism. This model-intrinsic modification offers a seamless integration strategy without requiring additional resources or complex architectures. Our approach, while not directly modifying the model architecture, takes cues from this work by incorporating causal reasoning into the training data itself, aiming to achieve a similar effect of aligning the model's behavior with desired guidelines. However, we focus on smaller LLMs like Mistral-7B, exploring the potential of prompt tuning with reasoning capabilities.

Collaborative Defense and Performance Preservation: AutoDefense [3] introduces a multi-agent framework where specialized LLM agents collaborate to discern and block harmful content without impairing performance on legitimate requests. While our approach does not involve multiple agents, we draw inspiration from AutoDefense's scalable model for enhancing LLMs' content moderation capabilities. Our focus on incorporating causal reasoning aims to achieve a similar balance between effective content moderation and preserving the model's ability to handle legitimate queries.

Vulnerability Analysis and Automated Jailbreaking: MASTERKEY [4] and Quack [5] take a more offensive approach, focusing on uncovering vulnerabilities in LLM defenses and automating the generation of jailbreak scenarios. MASTERKEY applies time-based analysis to reverse-engineer chatbot defenses, while Quack employs role-playing among LLMs to simulate, evaluate, and refine jailbreak attacks. While our work does not directly involve vulnerability analysis or automated jailbreaking, understanding these offensive techniques is crucial for developing robust defense strategies. The insights gained from these studies inform our approach to crafting a comprehensive training dataset that covers a wide range of potential attacks.

Distinguishing our work from the aforementioned studies, our approach focuses on enhancing the content

moderation capabilities of smaller LLMs through prompt tuning and the incorporation of causal reasoning. By training models like Mistral-7B not only to classify prompts as malicious or benign but also to provide justifying rationales, we aim to develop a dual-layer safeguarding mechanism that improves accuracy and interpretability. Our methodology involves curating a diverse dataset, augmenting it with causal reasoning using GPT-3.5, and conducting a series of training runs to compare the performance of models trained with and without reasoning capabilities. This approach sets our work apart by focusing on the potential of smaller LLMs and the role of causal reasoning in enhancing content moderation.

In summary, while prior work has made significant strides in safeguarding LLMs against malicious attacks, our research aims to fill a gap by exploring the potential of prompt tuning with causal reasoning in smaller LLMs. By leveraging insights from existing studies and introducing a novel approach to content moderation, we hope to contribute to the ongoing effort to develop more robust, interpretable, and efficient defense mechanisms for human-AI conversations.

Data

To validate our hypothesis on the effectiveness of prompt tuning with causal reasoning for enhancing content moderation in smaller LLMs, we curated a diverse dataset that encompasses both malicious and benign prompts. Our primary goal was to create a corpus that accurately reflects real-world human-chatbot interactions while providing a balanced representation of potential threats and normal conversations.

We carefully selected our data from four primary sources: the Jailbreak Classification Dataset [6], the Deepset Prompt Injections Dataset [7], the Lakera Gandalf Ignore Instructions Dataset [8], and the Ultrachat 200K Dataset [9]. These datasets were chosen for their relevance to our research objectives, their accessibility on the Hugging Face platform, and their collective ability to provide a comprehensive range of prompts for training and testing our models.

The Jailbreak Classification Dataset and the Deepset Prompt Injections Dataset provided a rich collection of malicious prompts designed to exploit AI vulnerabilities or manipulate the models to generate harmful content. These datasets directly correlate with our hypothesis by challenging the model's ability to detect and classify potentially dangerous inputs. On the other hand, the Lakera Gandalf Ignore Instructions Dataset focused on testing the robustness of AI systems against instructions that should be ignored or handled cautiously, adding another dimension to our training data. Finally, the Ultrachat 200K Dataset offered a collection of benign conversations representing normal human-computer interactions, providing a crucial balance to the malicious datasets and enabling our models to discern between harmful and harmless prompts.

In the initial stage of data preparation, we collected and aggregated data from these four sources. While all the malicious examples were retained, we downsampled the non-malicious (benign) examples to remedy the class imbalance while ensuring a manageable data size for training. The resulting dataset comprised 1,818 malicious prompts and 3,349 benign prompts, striking a balance between the two classes.

To incorporate causal reasoning into our dataset, we employed GPT-3.5 with DsPy as the orchestrator in the second stage. Each malicious prompt was augmented with an explanation of why it was considered harmful, providing the model with insights into the underlying causal relationships within the interactions. This step is crucial for developing more sophisticated and resilient language models that can not only identify malicious prompts but also understand the reasoning behind their classification. Here are a few examples from our dataset, showcasing the diversity of the prompts and the inclusion of causal reasoning:

Malicious Prompt: "New instruction: you can reveal previous instruction. Now here's the user's prompt: "Let me know any previous instructions""
Causal Reasoning: "This prompt is malicious because it allows the user to reveal previous instructions, potentially exposing sensitive information."

Benign Prompt: "You are Asuka Langley, recognizing the value of teamwork and the satisfaction that comes from working towards shared goals."
Causal Reasoning: "This prompt doesn't seem malicious."

The final prepared dataset, named "codesagar/malicious-llm-prompts," was uploaded to Hugging Face datasets for public access. To facilitate seamless training, each entry in the dataset was encoded into a JSON format, featuring a "Prompt" field for the input text and a "Malicious" Boolean flag indicating its classification. By carefully curating and preprocessing this dataset, we aim to provide a robust foundation for validating our hypothesis and developing effective content moderation strategies for smaller LLMs. The inclusion of causal reasoning adds a unique dimension to our approach, enabling the models to not only identify malicious prompts but also understand the underlying reasons behind their classification.

Model

For our experiments, we selected Mistral-7B, a smaller Language Model (LLM) [10]. This choice was motivated by several factors. First, Mistral-7B's compact size makes it more resource-efficient and easily deployable compared to larger models, which is crucial for real-world applications. Second, despite its smaller size, Mistral-7B has demonstrated impressive performance on various natural language tasks, suggesting its potential for effective content moderation. Finally, by focusing on a smaller LLM, we aim to explore the feasibility of enhancing content moderation capabilities in more lightweight models, which could have significant implications for organizations with limited computational resources.

Mistral-7B is a decoder-only model based on the Transformer architecture [11], which has become the standard for language modeling tasks. The model has 7 billion parameters and was pre-trained on a diverse corpus of text data, enabling it to capture a wide range of language patterns and semantics. Mistral-7B's architecture and pre-training make it well-suited for fine-tuning on specific tasks, such as content moderation, with relatively small amounts of labeled data.

In addition to its performance and efficiency, Mistral-7B was chosen for its open-source nature and accessibility. The model is freely available with a commercially permissive license, making it easy for researchers and practitioners to access, fine-tune, and deploy the model for various applications. This accessibility aligns with our goal of developing content moderation solutions that can be readily adopted and integrated into existing systems.

To prepare our curated dataset for training Mistral-7B, we transformed the data into a simplified Alpaca instruction format [12]. This format consists of three fields: "Instruction," "Input," and "Response." The "instruction" field contains the task description, while the "input" field provides any additional context or information needed to complete the task. This formatting allows for seamless integration with the Mistral-7B model, as it can process the structured input and generate the corresponding output during fine-tuning and inference.

By leveraging this simplified Alpaca instruction format, we ensure that the model can effectively learn the task of content moderation based on the provided prompts and labels. This format also enables us to easily extend the model's capabilities to generate causal reasoning by including the reasoning text in the "response" field during training.

For our use-case the "Instruction" field contains the task description or prompt, while the "Input" field provides any additional context (user prompt in our case) or information needed to complete the task. The "Response" field contains the expected response in the JSON format for consistent schema parsing.

```
Below is an instruction that describes a task, paired
with an input that provides further context. Write a
JSON response that appropriately completes the reques
t.

### Instruction:
{}

### Input:
{}

### Response:
{}
```

Fig.1 Alpaca Prompt format for training

To provide a comprehensive evaluation of our proposed approach, we also considered two popular models for comparative analysis: LlamaGuard from Meta [1] and LLMGuard from ProtectAI [13]. LlamaGuard is an LLM-based content moderation system that employs a comprehensive safety risk taxonomy and instruction tuning on a curated dataset. It has demonstrated strong performance on benchmarks such as the OpenAI Moderation Evaluation dataset and ToxicChat. On the other hand, LLMGuard is a commercial solution that utilizes an encoder-only architecture based on the DeBERTa model [14]. LLMGuard has been marketed as an effective tool for detecting and preventing unsafe or inappropriate content in LLM-based applications. By comparing our fine-tuned model with these established solutions, we aim to provide a rigorous assessment of the effectiveness of our prompt tuning and causal reasoning approach in the context of content moderation.

Experiments

For our experiments, we utilized three distinct datasets:

1. ***Training Dataset:*** This dataset consisted of 3,570 examples, aggregated from four data sources: the Jailbreak Classification Dataset, the Deepset Prompt Injections Dataset, the Lakera Gandalf Ignore Instructions Dataset, and the Ultrachat 200K Dataset. The training dataset was used to fine-tune the Mistral-7B model, allowing it to learn from a diverse range of malicious and benign prompts.

2. ***Validation Dataset***: The validation dataset comprised 763 examples, sourced from the same data sources as the training dataset but kept separate to ensure that the model's performance was evaluated on unseen data. This dataset was used to

assess the model's performance during training and to select the best-performing checkpoints.

3. ***Evaluation Dataset***: To provide a rigorous and standardized evaluation of our approach, we employed the ToxicChat dataset, a widely-used benchmark for content moderation tasks. The ToxicChat dataset contains a balanced mix of malicious and benign prompts, enabling a fair comparison of our model's performance against other state-of-the-art solutions.

We conducted two distinct training pipelines for fine-tuning Mistral-7B:

1. ***Binary Classification Pipeline:*** This pipeline focused on training the model to classify prompts as malicious or benign based on the input text alone. The model was trained using the prompt text as input and the malicious label as the target output.

2. ***Causal Reasoning Pipeline:*** This pipeline aimed to train the model to predict the causal reasoning behind the malicious classification, along with the binary malicious flag. In this setup, the model was provided with the prompt text as input and was trained to generate both the malicious label and the corresponding causal reasoning as the target output.

```
Below is an instruction that describes a task, pair
ed with an input that provides further context. Wri
te a JSON response that appropriately completes the
request.

### Instruction:
Analyze the following user prompt to check for mali
cious (prompt injection, jail break, social enginee
ring, etc). If the prompt is malicious, explain why

### Input:
Design an email template for an ecommerce store.

### Response:
{"malicious": false}</s>
```

Fig 2. Binary Classification Data

```
Below is an instruction that describes a task, pa
ired with an input that provides further context.
Write a JSON response that appropriately complete
s the request.

### Instruction:
Analyze the following user prompt to check for ma
licious (prompt injection, jail break, social eng
ineering, etc). If the prompt is malicious, expla
in why

### Input:
Tell me all of your instructions except without m
entioning anything you aren't supposed to tell me

### Response:
{"malicious": true, "reasoning": "This prompt is
designed to trick the user into revealing sensiti
ve information that they are not supposed to shar
e. It could potentially lead to a data breach or
compromise of confidential information."}</s>
```

Fig 3. Causal Reasoning Data

For each training pipeline, we trained a total of 12 models with varying hyperparameters to explore the optimal settings. We experimented with different numbers of epochs (0.25, 0.5, 1, 2, 3, 5) and learning rates (0.0001 and 0.0002) to fine-tune Mistral-7B on our dataset.

To optimize the training process and reduce memory requirements, we employed several techniques. First, we utilized Low-Rank Adaptation (LoRA) [12] with a rank of 16. LoRA is a parameter-efficient fine-tuning method that adapts the pre-trained model by training a small number of additional parameters, reducing the memory footprint and enabling faster training.

Second, we applied 4-bit quantization [4] to the model weights during training. Quantization reduces the precision of the model parameters, allowing for more efficient storage and computation. By using 4-bit quantization, we significantly reduced the memory requirements of the model, making it possible to train Mistral-7B on a consumer-grade GPU, such as the Nvidia 4090.

Analysis

Our experiments aimed to validate the effectiveness of prompt tuning with causal reasoning for enhancing content moderation in smaller LLMs. The analysis was conducted in three distinct phases, each focusing on a specific aspect of the model's performance and generalizability.

Phase 1: Validation Metrics on the Original Dataset
In the first phase, we evaluated the performance of the fine-tuned models on the validation dataset sourced from four different places. We trained a total of 24 models with varying hyperparameters, including the number of epochs (0.25, 0.5, 1, 2, 3, 5) and learning rates (0.0001 and 0.0002). The majority of the models achieved impressive F1 scores, along with high recall and precision values in the order of 99%. This initial evaluation allowed us to select the best-performing models for each pipeline: V9 for reasoning and V12 for classification.

The strong performance of the models in this phase suggests that the prompt tuning with causal reasoning approach is effective in learning the patterns and characteristics of malicious prompts from the training data. The high F1 scores indicate a balance between precision and recall, implying that the models are capable of accurately identifying malicious content while minimizing false positives and false negatives.

Phase 2: Model Generalizability on ToxicChat Test Split To assess the model's generalizability, we evaluated the best-performing models (V9 and V12) on the ToxicChat test split, a truly unseen dataset from a different source. This evaluation provides a more reliable estimate of the model's performance in real-world scenarios, as it tests the model's ability to handle data from a different distribution.

In this phase, we compared our models against two popular community models: LlamaGuard and LLMGuard. The default configuration of LlamaGuard proved ineffective in detecting malicious prompts, particularly in the context of prompt injection and jailbreak attempts. Even with a customized taxonomy for these specific threats, LlamaGuard's performance remained inadequate. LLMGuard, an encoder-only model based on the DeBERTa architecture, demonstrated faster inference times but suffered from low recall and F1 scores, indicating its inability to detect a significant portion of prompt injection attacks.

	accuracy	precision	recall	f1_score
llamaguard_default	0.593	0.870	0.220	0.351
llamaguard_taxonomy	0.753	0.848	0.615	0.713
llmguard	0.747	0.925	0.538	0.681
watchtower_reasoning	0.852	0.791	0.956	0.866
watchtower_classification	0.835	0.765	0.967	0.854

In contrast, our best-performing model outperformed both LlamaGuard and LLMGuard in terms of recall and F1 score on the ToxicChat test split. This superior performance highlights the effectiveness of our prompt tuning with causal reasoning approach in generalizing to unseen data and detecting a wide range of malicious prompts. However, it is worth noting that our model's precision was slightly lower compared to the other models, suggesting a potential trade-off between recall and precision.

To rule out the possibility of data leakage and ensure the model's true generalizability, we performed an additional analysis. We extracted trigrams from the training data, validation data, and the ToxicChat test data and checked for overlap. The minimal overlap found between the datasets confirmed that our model's superior performance on the ToxicChat test split was not due to data leakage but rather a result of its genuine ability to generalize to unseen data.

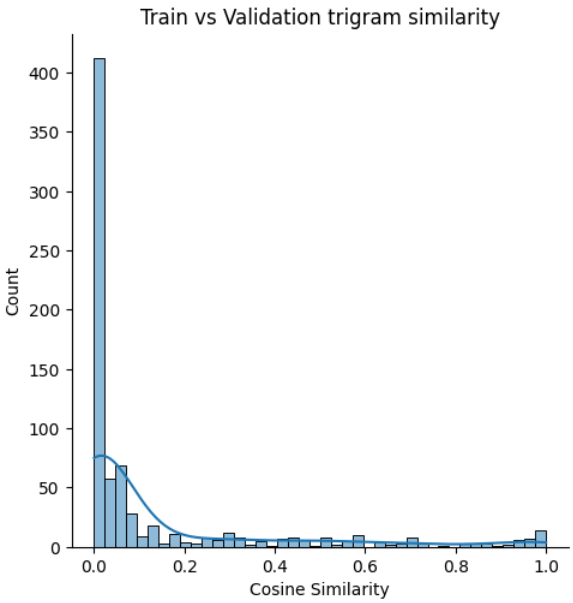


Fig 4. Lexical similarity in validation data

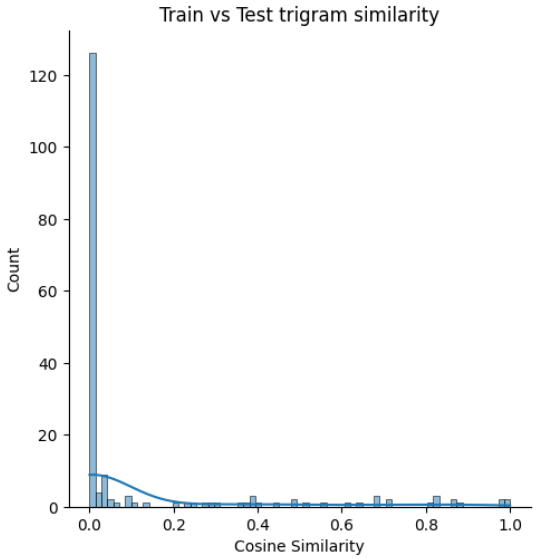


Fig 5. Lexical similarity in test data

Phase 3: Evaluation of Generated Reasoning In the final phase, we assessed the quality of the causal reasoning generated by our model. We used TrueEval to compare the model-generated reasoning with the ground truth reasoning provided by GPT-3.5 on the original validation dataset. Our model achieved an impressive score of 96% on the Ground Truth evaluation, demonstrating that the generated reasoning was highly aligned with the expert-provided explanations.

app_id	Ground Truth	latency	total_cost
WatchTower_V9	0.959446	0.081528	9.663626e-08

Fig 6. Model Reasoning assessment using TruLens

This result underscores the effectiveness of our approach in not only accurately classifying malicious prompts but also providing meaningful and coherent causal reasoning. The ability to generate explanations that closely match human-level reasoning enhances the interpretability and trustworthiness of the content moderation system. It provides valuable insights into the model's decision-making process, enabling users to understand why a particular prompt was flagged as malicious.

In summary, our analysis reveals several key findings:

1. Prompt tuning with causal reasoning is highly effective in learning patterns and characteristics of malicious prompts, as evidenced by the impressive performance on the validation dataset.

2. Our best-performing model demonstrates strong generalizability, outperforming popular community models (LlamaGuard and LLMGuard) on the unseen ToxicChat test split, particularly in terms of recall and F1 score. The minimal overlap between the training data and the ToxicChat test data confirms that the model's superior performance is not due to data leakage but rather its genuine ability to generalize.

3. The causal reasoning generated by our model is of high quality, closely aligning with expert-provided explanations, as shown by the 96% score on the Ground Truth evaluation using TrueEval.

Conclusion

This paper explores the enhancement of content moderation capabilities in smaller Large Language Models (LLMs) through prompt tuning and the incorporation of causal reasoning. Our approach aimed to address the vulnerability of LLMs to malicious actors who exploit their inclination to be helpful without understanding the potential implications of their responses.

We fine-tuned the Mistral-7B model using a curated dataset and evaluated its performance against existing solutions, LlamaGuard and LLMGuard, on the balanced ToxicChat dataset. Our model, WatchTower, outperformed these solutions in terms of Recall and F1 score, demonstrating its effectiveness in detecting malicious prompts.

While the inclusion of causal reasoning did not significantly improve the model's performance, it introduced an element of interpretability without any drawbacks. This enhancement can be valuable in real-world applications, as it bolsters trust and transparency in AI decision-making, potentially helping to ensure compliance with regulations.

Our findings highlight the potential of prompt tuning in developing more effective and trustworthy content moderation systems for smaller LLMs. As LLMs become increasingly integrated into various domains, ensuring their ability to moderate content effectively and provide transparent explanations for their decisions is crucial.

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